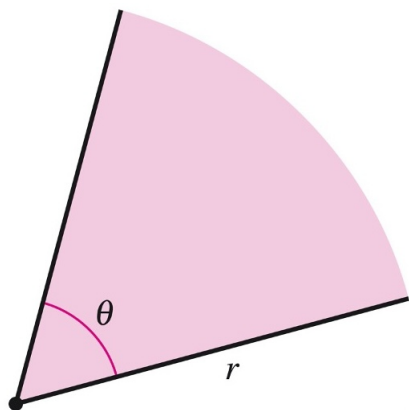


Math 252, 10.5, Area Polar Regions

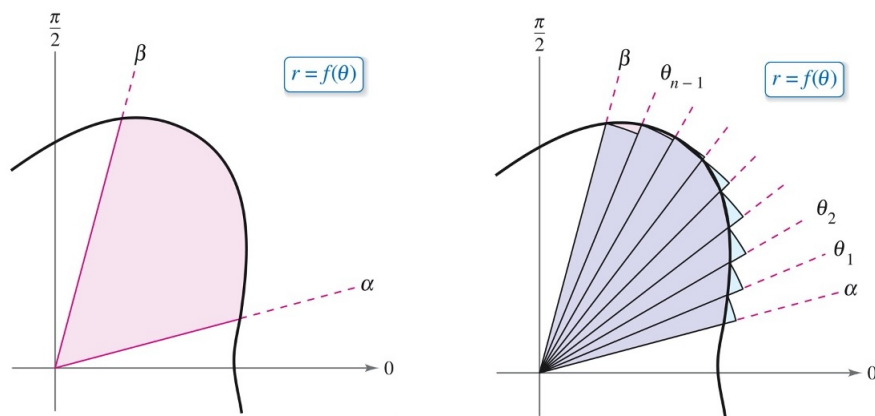
1. **Area of a Sector** A sector of a circle is a region of the circle subtended by an angle θ



Since the sector represents the fraction $\frac{\theta}{2\pi}$ of the entire circle,

$$\begin{aligned} A_{\text{sector}} &= \frac{\theta}{2\pi} (\pi r^2) \\ &= \frac{1}{2} r^2 \theta. \end{aligned}$$

2. **Polar Regions.** Consider the function $r = f(\theta)$, for $\alpha < \theta < \beta$.



Partition the interval $[\alpha, \beta]$ into n subintervals of width

$$\Delta\theta = \frac{\beta - \alpha}{n}.$$

On the i th subinterval, choose a sample angle θ_i^* . The corresponding thin polar region is approximately a sector with radius $f(\theta_i^*)$ and central angle $\Delta\theta$, so

$$\Delta A_i \approx \frac{1}{2} [f(\theta_i^*)]^2 \Delta\theta.$$

Adding the areas of the sectors gives

$$A \approx \sum_{i=1}^n \frac{1}{2} [f(\theta_i^*)]^2 \Delta\theta.$$

Taking the limit as $n \rightarrow \infty$ gives the Riemann sum

$$\begin{aligned} A &= \lim_{n \rightarrow \infty} \sum_{i=1}^n \frac{1}{2} [f(\theta_i^*)]^2 \Delta\theta \\ &= \int_{\alpha}^{\beta} \frac{1}{2} [f(\theta)]^2 d\theta. \end{aligned}$$

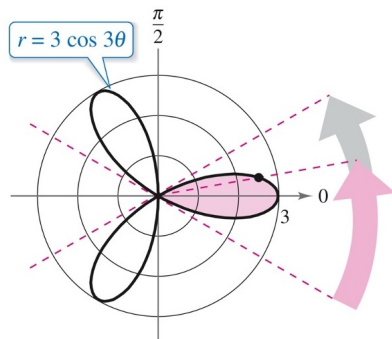
3. **Area of a Polar Region..** If f is continuous and nonnegative on the interval $[\alpha, \beta]$, $0 < \beta - \alpha < 2\pi$, then the area of the region of the area bounded between α and β is given by,

$$\int_{\alpha}^{\beta} \frac{1}{2} [f(\theta)]^2 d\theta$$

4. **Main Tool for Integration..** We use the half-angle formulas from trigonometry

$$\cos^2 \theta = \frac{\cos 2\theta + 1}{2} \quad \sin^2 \theta = \frac{1 - \cos 2\theta}{2}$$

5. **Example 1:** Determine the area one petal of the graph of $r = 3 \cos 3\theta$.



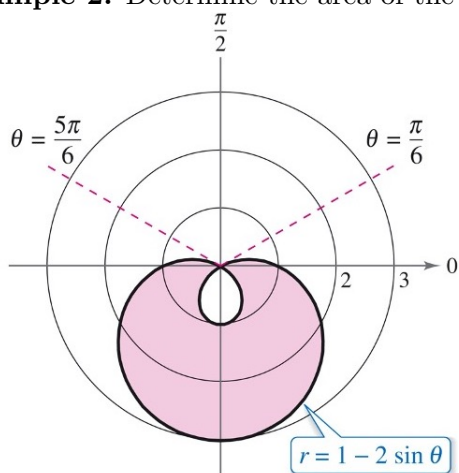
The petal centered on the polar axis begins and ends where $r = 0$:

$$3 \cos 3\theta = 0 \implies \theta = -\frac{\pi}{6}, \frac{\pi}{6}.$$

Therefore,

$$\begin{aligned} A &= \frac{1}{2} \int_{-\pi/6}^{\pi/6} (3 \cos 3\theta)^2 d\theta \\ &= \frac{9}{2} \int_{-\pi/6}^{\pi/6} \cos^2 3\theta d\theta \\ &= \frac{9}{2} \int_{-\pi/6}^{\pi/6} \frac{1 + \cos 6\theta}{2} d\theta \\ &= \frac{9}{4} \left[\theta + \frac{1}{6} \sin 6\theta \right]_{-\pi/6}^{\pi/6} \\ &= \boxed{\frac{3\pi}{4}}. \end{aligned}$$

6. **Example 2:** Determine the area of the outer loop of the limaçon with inner loop given by $r = 1 - 2 \sin \theta$.



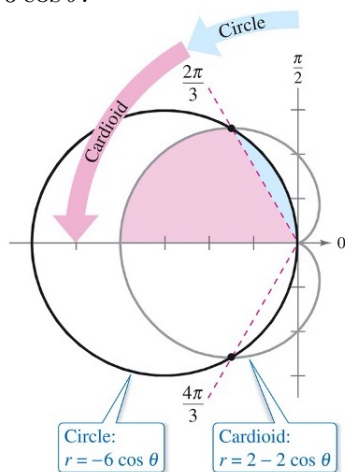
First determine where the curve passes through the pole:

$$\begin{aligned} 1 - 2 \sin \theta &= 0 \\ \sin \theta &= \frac{1}{2}, \end{aligned}$$

so $\theta = \frac{\pi}{6}$ and $\theta = \frac{5\pi}{6}$. The outer loop is traced from $\theta = \frac{5\pi}{6}$ to $\theta = \frac{13\pi}{6}$. Thus,

$$\begin{aligned} A &= \frac{1}{2} \int_{5\pi/6}^{13\pi/6} (1 - 2 \sin \theta)^2 d\theta \\ &= \frac{1}{2} \int_{5\pi/6}^{13\pi/6} (1 - 4 \sin \theta + 4 \sin^2 \theta) d\theta \\ &= \frac{1}{2} \int_{5\pi/6}^{13\pi/6} (3 - 4 \sin \theta - 2 \cos 2\theta) d\theta \\ &= \frac{1}{2} [3\theta + 4 \cos \theta - \sin 2\theta]_{5\pi/6}^{13\pi/6} \\ &= \boxed{2\pi + \frac{3\sqrt{3}}{2}}. \end{aligned}$$

7. **Example 3:** Determine the area of the region bounded between the cardioid $r = 2 - 2 \cos \theta$ and the circle $r = -6 \cos \theta$.



Find the angles at which the curves intersect:

$$\begin{aligned} 2 - 2 \cos \theta &= -6 \cos \theta \\ 2 &= -4 \cos \theta \\ \cos \theta &= -\frac{1}{2}. \end{aligned}$$

Thus, the curves intersect at $\theta = \frac{2\pi}{3}$ and $\theta = \frac{4\pi}{3}$. By symmetry about the polar axis, we may find the area above the axis and double it. On $\left[\frac{\pi}{2}, \frac{2\pi}{3}\right]$, the circle is the inner curve, and on $\left[\frac{2\pi}{3}, \pi\right]$, the cardioid is the inner curve. Therefore,

$$\begin{aligned} A &= 2 \left[\frac{1}{2} \int_{\pi/2}^{2\pi/3} (-6 \cos \theta)^2 d\theta + \frac{1}{2} \int_{2\pi/3}^{\pi} (2 - 2 \cos \theta)^2 d\theta \right] \\ &= \int_{\pi/2}^{2\pi/3} 36 \cos^2 \theta d\theta + \int_{2\pi/3}^{\pi} (2 - 2 \cos \theta)^2 d\theta \\ &= 2 \left[\left(\frac{3\pi}{2} - \frac{9\sqrt{3}}{4} \right) + \left(\pi + \frac{9\sqrt{3}}{4} \right) \right] \\ &= \boxed{5\pi}. \end{aligned}$$